

COURSE 10: GENERATIVE AI – ENHANCE YOUR DATA ANALYTICS CAREER

MODULE 1 DETAILED NOTES: DATA ANALYTICS AND GENERATIVE AI

1.1 INTRODUCTION TO GENERATIVE AI IN DATA ANALYTICS

1.1.1 What is Generative AI?

Generative AI is a branch of Artificial Intelligence that focuses on creating new content or synthetic data rather than simply classifying or predicting outcomes.

Unlike traditional AI models:

Traditional AI	Generative AI
Predicts outcomes	Creates new content
Classifies data	Generates data
Detects patterns	Produces text, images, code, audio, etc.

Examples

- Generate synthetic datasets
- Fill missing values
- Generate SQL queries
- Create visualizations
- Build dashboards
- Produce reports and narratives
- Generate code from natural language prompts

1.1.2 Importance of Generative AI in Data Analytics

Generative AI is transforming the entire analytics lifecycle.

Traditional Analytics Process

1. Data Collection
2. Data Cleaning
3. Data Preparation
4. Data Analysis
5. Visualization
6. Reporting
7. Decision Making

Generative AI Enhancements

- Faster data preparation
- Automated analysis
- Natural language querying
- Automated reporting
- Interactive dashboards
- Storytelling from data

1.1.3 Applications of Generative AI in Data Analytics

A. Data Generation

Generative AI can create synthetic datasets when real-world data is:

- Limited
- Incomplete
- Imbalanced
- Sensitive

Benefits

- Improves model training
- Preserves privacy
- Simulates scenarios
- Reduces data scarcity

Example

Generate a patient dataset containing:

- Age
- Blood Sugar
- Symptoms
- Diagnosis

B. Data Augmentation

Data augmentation means artificially increasing dataset size.

Benefits

- Improves machine learning accuracy
- Handles class imbalance
- Creates diverse training samples

Types

1. Structured Data

Examples:

- Tables
- CSV files

Tools:

- CTGAN
- SDV (Synthetic Data Vault)

2. Semi-Structured Data

Examples:

- Text
- Code

Tools:

- GPT
- GitHub Copilot

3. Unstructured Data

Examples:

- Images
- Audio
- Videos

Tools:

- StyleGAN2
- BigGAN
- SoundGAN

C. Data Imputation

Generative AI can:

- Detect missing values
- Fill missing values intelligently
- Improve data quality

Example

Before:

Temperature

70
NULL

Temperature

72

After:

Temperature

70
71
72

D. Data Cleaning

Automates:

- Error detection
- Duplicate removal
- Standardization
- Formatting

Example

Converting:

USA
U.S.A.
United States

into:

United States

E. Data Normalization

Transforms data into consistent scales.

Example

Normalize CPU frequency:

2.5 GHz
3.0 GHz
4.0 GHz

Using Max Normalization:

0.625 0.75 1.0

F. Data Transformation

Generative AI can:

- Convert text to image
- Convert image to text
- Convert code to documentation
- Convert datasets into summaries

1.2 GENERATIVE AI FOR DATABASE QUERYING

Traditional SQL querying requires technical expertise.

Generative AI allows:

Natural Language Queries

Example:

Instead of:

SELECT SUM(Sales) FROM Orders WHERE Year=2024;
--

Ask:

What were the total sales in 2024?

Generative AI converts it into SQL automatically.

Benefits

- Faster querying
- Less SQL knowledge required
- Real-time responses
- Reduced errors

1.3 GENERATIVE AI FOR QUESTION ANSWERING (Q&A)

Users can interact with data conversationally.

Example

Question:

Which product generated the highest revenue?
--

AI Response:

Product A generated ₹25 million revenue in 2024.
--

Technologies

1. GPT

- Natural language generation
- Conversational analytics

2. BERT

- Context understanding
- Query interpretation

1.4 GENERATIVE AI FOR DATA VISUALIZATION

Visualization converts raw data into meaningful visuals.

Generative AI helps by:

- Creating charts automatically
- Selecting best chart types
- Generating dashboards
- Enhancing aesthetics

Examples of Visualization Tools

1. Tableau AI

- Natural language insights
- Auto-generated charts

2. IBM Cognos Analytics

- AI-assisted reporting
- Insight generation

3. Google Looker AI

- Interactive analytics
- AI-powered exploration

Benefits

- Better understanding
 - Faster reporting
 - Improved communication
 - Interactive exploration
-

1.5 GENERATIVE AI FOR DASHBOARD CREATION

Generative AI can automatically:

- Design layouts
- Add widgets
- Generate KPIs
- Personalize dashboards

Advantages

- Faster dashboard development
 - Better user experience
 - Dynamic content generation
-

1.6 GENERATIVE AI FOR DATA STORYTELLING

Data storytelling converts analysis into narratives.

Generative AI helps by:

Generating:

- Reports
- Executive summaries
- Recommendations
- Insights

Example

Raw Data:

Sales increased by 15%

Generated Story:

Sales increased by 15% during Q4, driven primarily by increased demand in urban markets and successful promotional campaigns.

1.7 POPULAR GENERATIVE AI TOOLS FOR DATA ANALYTICS

Analytics Platforms

1. DataRobot

- AutoML
- Model building
- Predictive analytics

2. RapidMiner

- Data science platform
- Machine learning

3. Tableau AI

- Visualization
- Dashboard generation

4. IBM Cognos Analytics

- AI-driven analytics

Generative Models

1. DALL-E 2

- Text-to-image generation

2. Google Imagen

- Image generation

3. StyleGAN2

- Face generation

4. BigGAN

- High-resolution image generation

5. CycleGAN

- Style transfer

NLP Tools

1. Hugging Face Transformers

- GPT
- BART
- Language generation

2. spaCy

- NLP processing

3. Stanford CoreNLP

- Language analysis

Explainable AI Tools

1. LIME

Explains ML model predictions.

2. SHAP

Measures feature importance.

3. Captum

Explainable AI library for PyTorch.

Data Preparation Tools

1. DataWrangler

- Cleaning
- Transformation

2. OpenRefine

- Data normalization
- Data transformation

Feature Engineering Tools

1. Featuretools

Automatic feature creation.

2. H2O Driverless AI

Automatic feature selection.

3. TPOT

Automated ML pipeline generation.

1.8 REAL-WORLD INDUSTRY EXAMPLES

1. Alteryx AiDIN

Used for:

- Enterprise analytics
- Decision making
- Predictive recommendations

Example

PGA golfer Martin Trainer uses Alteryx for performance recommendations.

2. Netmeds + Haptik

AI chatbot implementation.

Results

- Saved 2600+ agent hours
- 83.6% automation rate
- 99% improvement in response time

3. DataRobot

Used for:

- Synthetic data generation
- Model building
- Manufacturing optimization

4. Palantir AIP

Provides:

- Secure LLM deployment
- Data governance
- AI integration

5. Google Analytics 4

Uses AI for:

- Predictive audiences
- Automated insights
- Customer behavior prediction

Example

McDonald's Hong Kong used GA4 for ecommerce analytics.

6. Facebook Business Suite

Uses AI for:

- Audience insights
- Ad optimization
- Content generation
- Predictive analytics

Example

Vision Express saved:

- 36 hours/month
 - Doubled customer inquiries
-

1.9 IBM WATSONX CASE STUDY

What is Watsonx?

IBM Watsonx is an enterprise AI and analytics platform.

Capabilities

1. Data Collection

- Databases
- Cloud storage
- Real-time streams

2. Data Preparation

- Cleaning
- Wrangling
- Transformation

3. Exploratory Data Analysis

- Visual exploration
- Pattern detection

4. Machine Learning

- Predictive analytics
- Forecasting

5. Real-Time Analytics

- Instant insights

6. Explainable AI

- Transparent decision making
-

1.10 NATWEST MORTGAGE SUPPORT CASE STUDY

Problem

Mortgage applications were becoming increasingly complex.

Customers needed:

- Accurate information
- Faster support

Solution

IBM + NatWest created:

"Marge"

AI-powered mortgage assistant built using:

- IBM Watson Assistant
- IBM Cloud

Results

Customer Satisfaction

- 20% increase in Net Promoter Score (NPS)

Efficiency

- 10% reduction in call duration

Benefits

- Real-time support
 - Faster information retrieval
 - Improved employee productivity
-

1.11 HANDS-ON LAB: DATAROBOT

Objectives

Learn to:

- Create a DataRobot account
- Build use cases
- Upload datasets
- Train models
- Generate predictions

Workflow

Step 1

Create account

Step 2

Create Use Case

Step 3

Add Dataset

Example:

- HOSPITAL_READMISSION_DATA

Step 4

Choose Target Variable

Example:

- Readmitted

Step 5

Train Models

Step 6

Review:

- Feature Importance
- Blueprint
- Model Overview

Step 7

Generate Predictions

Step 8

Download Compliance Reports

1.12 KEY BENEFITS OF GENERATIVE AI IN DATA ANALYTICS

1. Improved Data Quality

- Missing value handling
- Error correction

2. Better Model Performance

- Data augmentation
- Synthetic data generation

3. Faster Analytics

- Automated workflows

4. Hidden Pattern Discovery

- Advanced pattern detection

5. Personalized Insights

- Context-aware recommendations

6. Better Storytelling

- Automated narratives

7. Ethical Considerations

- **Data Privacy:** Sensitive data must be protected.
 - **Concern:** Generative AI may expose confidential information.
 - **Security Risks:** According to surveys, many IT leaders worry about:
 1. Data leakage
 2. Unauthorized access
 3. Model misuse
 - **Transparency:** Organizations should:
 4. Explain AI decisions
 5. Ensure accountability
 - **Fairness:** AI systems should avoid:
 6. Bias
 7. Discrimination
 8. Unfair outcomes
-

1.13 GENERATIVE AI FOR QUERYING DATABASES

What is Database Querying?

A **database** is an organized collection of data stored electronically.

A **query** is a request made to a database to:

- Retrieve data
- Update data
- Delete data
- Insert data
- Analyze data

The most commonly used language for querying relational databases is:

SQL (Structured Query Language)

SQL allows users to:

- Select data
- Filter records
- Sort information
- Aggregate values
- Update records
- Create tables

Example:

```
SELECT * FROM Employees;
```

This retrieves all employee records.

Role of Generative AI in Database Querying

Traditionally:

- Users needed SQL knowledge.
- Query creation could take significant time.
- Complex databases required understanding schema structures.

With Generative AI:

Users can ask questions in natural language.

Example:

Natural Language

```
"Show all employees earning more than ₹50,000."
```

Generative AI converts it into:

```
SELECT *  
FROM Employees  
WHERE Salary > 50000;
```

Benefits of AI-Powered Database Querying

1. Natural Language Processing (NLP)

Allows users to communicate using everyday language.

Example:

"How many customers purchased products last month?"

AI generates:

SELECT COUNT(*)
FROM Customers
WHERE PurchaseDate >= '2026-05-01';

2. Faster Query Generation

Instead of manually writing SQL:

- Describe requirement
- AI creates query instantly

3. Reduced Learning Curve

Non-technical users can interact with databases without SQL expertise.

4. Improved Productivity

Data professionals spend less time:

- Writing queries
- Debugging syntax
- Understanding database structures

and more time:

- Interpreting insights
- Making decisions

SQL Through AI Demonstration

Dataset Used:

Boston Housing Price Dataset

The AI tool converts natural language requests into SQL commands.

Example 1: Find Column Names

Prompt:

What are the column names?

Generated SQL:

```
SHOW COLUMNS FROM Boston_house_prices;
```

Purpose:

Displays all attributes in the table.

Example 2: Count Number of Rows

Prompt:

Count the rows in the dataset.

Generated SQL:

```
SELECT COUNT(*)  
FROM Boston_house_prices;
```

Purpose:

Returns total records.

Example 3: Average Age

Prompt:

Find average age.

Generated SQL:

```
SELECT AVG(age)  
FROM Boston_house_prices;
```

Purpose:

Calculates average age.

Example 4: Filter Rows Based on Tax Range

Prompt:

Find rows where tax is between 210 and 250.

Generated SQL:

```
SELECT *  
FROM Boston_house_prices  
WHERE tax >= 210  
AND tax <= 250;
```

Purpose:

Retrieves matching records.

Example 5: Replace Values

Prompt:

Replace ZN value 0 with 5.

Generated SQL:

```
UPDATE Boston_house_prices  
SET ZN = 5  
WHERE ZN = 0;
```

Purpose:

Updates records.

Example 6: Sort Table

Prompt:

Sort by MEDV ascending.

Generated SQL:

```
SELECT *  
FROM Boston_house_prices  
ORDER BY MEDV ASC;
```

Purpose:

Sorts data from lowest to highest.

Example 7: Insert New Rows

Prompt:

Insert new row.

Generated SQL:

```
INSERT INTO Boston_house_prices  
VALUES (...);
```

Purpose:

Adds new records.

Example 8: Multiple Conditions

Prompt:

Find rows where RAD = 5 and age between 50 and 55.

Generated SQL:

```
SELECT *  
FROM Boston_house_prices  
WHERE RAD = 5  
AND age BETWEEN 50 AND 55;
```

Example 9: Create Sub-table

Prompt:

Create sub-table where CHAS = 1 and RAD = 4.

Generated SQL:

```
SELECT *  
FROM Boston_house_prices  
WHERE CHAS = 1  
AND RAD = 4;
```

1.14 HANDS-ON LAB: QUERYING HEART DISEASE DATASET

The lab demonstrates how providing context improves query quality.

Dataset Attributes

Key attributes include:

Attribute	Description
age	Age in years
gender	Male/Female
cp	Chest pain type
trestbps	Blood pressure
chol	Cholesterol
fbs	Blood sugar
restecg	ECG results
thalach	Maximum heart rate
exang	Exercise-induced angina
oldpeak	ST depression
slope	ST segment slope
ca	Major vessels count
thal	Thallium test result
num	Heart disease status

Sample SQL Queries Generated

Age Distribution

```
SELECT
MIN(age) AS min_age,
MAX(age) AS max_age,
AVG(age) AS avg_age
FROM heart_disease_prediction_dataset;
```

Gender Analysis

```
SELECT
SUM(CASE WHEN gender=1 THEN 1 ELSE 0 END) AS male_count,
SUM(CASE WHEN gender=0 THEN 1 ELSE 0 END) AS female_count
FROM heart_disease_prediction_dataset;
```

Chest Pain Frequency

```
SELECT cp,
COUNT(*)
FROM heart_disease_prediction_dataset
GROUP BY cp;
```

Purpose:

Determines most common chest pain type.

Age Group vs Heart Disease

Groups patients by age range and analyzes disease prevalence.

Useful for:

- Risk assessment
- Population analysis
- Healthcare planning

Important Practice Prompts

Generative AI can create queries for:

Cholesterol Range

```
SELECT  
MIN(chol),  
MAX(chol)  
FROM heart_disease_prediction_dataset;
```

Male/Female Age Range

```
SELECT gender,  
MIN(age),  
MAX(age)  
FROM heart_disease_prediction_dataset  
GROUP BY gender;
```

Maximum Heart Rate by Age Group

```
SELECT age_group,  
MAX(thalach)  
FROM ...
```

Percentage with High Blood Sugar

```
SELECT  
(COUNT(CASE WHEN fbs=1 THEN 1 END)*100.0)/COUNT(*)  
AS percentage  
FROM heart_disease_prediction_dataset;
```

Patients with Reversible Thalassemia

```
SELECT COUNT(*)  
FROM heart_disease_prediction_dataset  
WHERE thal=7;
```

1.15 Q&A FOR DATA (QUESTION AND ANSWER SYSTEMS)

Definition

Q&A for Data refers to:

Asking questions about data using natural language and receiving meaningful answers.

Instead of writing SQL, users simply ask:

- What were total sales last month?
- Which product sold the most?
- Which region generated highest revenue?

The system analyzes data and provides answers.

Advantages of Q&A Systems

1. Data Exploration

Users can investigate datasets easily.

2. Insight Generation

Helps discover:

- Trends
- Patterns
- Correlations
- Outliers

3. Accessibility

Even non-technical users can analyze data.

4. Faster Decision-Making

Immediate answers support business decisions.

Generative AI Frameworks for Data Q&A

1. IBM Cognos Analytics

Features:

- AI-assisted data discovery

- Automated pattern detection
- Natural language querying
- Dashboard generation

Benefits:

- Simplifies business intelligence
- Generates reliable insights

2. Microsoft Power BI

Features:

- Ask questions in natural language
- Interactive data exploration
- Dashboard integration
- AI-powered visualizations

Example:

"Show monthly sales trend."

Power BI automatically creates charts.

3. Tableau Pulse

Features:

- Personalized insights
- Context-aware recommendations
- Suggested follow-up questions
- Automated analytics

Benefits:

- Makes organizations data-driven.

4. ThoughtSpot Sage

Features:

- LLM-powered search
- Natural language analytics

- Personalized recommendations
- Suggested synonyms

Benefits:

- Easy for both technical and non-technical users.

5. Sisense Analytics Chatbot

Features:

- Conversational analytics
- Visualization generation
- Narrative explanations
- Sample insights

Benefits:

- Single chat interface for data exploration.

6. Dash Enterprise

Features:

- Smart Insights
- Anomaly detection
- Outlier identification
- Automated analysis

Benefits:

- Quickly extracts business insights.

7. Crystal (iGenius)

Features:

- Conversational AI
- KPI monitoring
- Personalized alerts
- Trend analysis
- Data storytelling

Benefits:

- Connects business teams and data teams.
-

1.16 HANDS-ON LAB: POWERDRILL AI

Purpose

Powerdrill enables users to upload datasets and interact using natural language.

Website:

- Upload dataset
- Ask questions
- Receive insights and visualizations

Example Questions

Top Selling Products

Prompt:

What are the top-selling products?

Output:

Identifies highest-selling products.

Highest Average Price

Prompt:

Which product has the highest average price?

Output:

Shows premium products.

Total Revenue

Prompt:

What is the total revenue for the latest two months?

Output:

Revenue summary.

Quantity Ordered Over Time

Prompt:

How has the quantity ordered changed over time?

Output:

Trend analysis.

Distribution of Orders

Prompt:

What is the distribution of order quantities?

Output:

Histogram showing order frequency.

Correlation Analysis

Prompt:

Are there correlations between quantity ordered and price each?

Output:

Relationship between price and demand.

Highest Sales Day

Prompt:

Which day of the week has the highest sales?

Output:

Identifies most profitable day.

Average Order Size

Prompt:

What is the average order size?

Output:

Typical customer purchase volume.

1.17 SUCCESSFUL IMPLEMENTATIONS OF GENERATIVE AI

Generative AI is widely used for:

- Data Preparation
- Data Augmentation
- Synthetic Data Generation
- Business Intelligence
- Decision Support

Industry Applications

1. Healthcare

Applications:

- Synthetic medical images
- Patient records
- Drug discovery
- Clinical research

Benefits:

- Better model training
- Improved diagnostics

2. Finance

Applications:

- Risk modeling
- Fraud detection
- Anomaly detection

Benefits:

- Improved financial forecasting
- Better fraud prevention

3. Retail

Applications:

- Product descriptions
- Product images
- Recommendation systems
- Personalized marketing

Benefits:

- Better customer engagement

Real-World Company Examples

1. Walmart – My Assistant

Introduced in 2023.

Used by:

- 50,000+ corporate employees

Functions:

- Summarization
- Content generation
- Data synthesis

Benefits:

- Increased employee productivity

2. OpenAI

GPT models help with:

- Data augmentation
- Data quality improvement
- Synthetic data generation

3. Typeform (Formless)

Features:

- Conversational forms
- Multi-language support
- AI-generated follow-ups

Benefits:

- Better data collection

4. Stanford University

Research Project:

RoentGen

Uses diffusion models to generate:

- Clinical chest X-ray images

Benefits:

- Improves medical AI training
- Supports rare disease diagnosis

5. Sigma

Features:

- Spreadsheet-like interface
- Data quality correction
- Cloud warehouse integration

Example:

Used by Workato.

6. DataRobot

Uses Generative AI for:

- Data cleaning
- Feature engineering
- Data transformation

7. H2O.ai

Platform:

Driverless AI

Uses:

- GANs (Generative Adversarial Networks)

Functions:

- Automated data preparation
 - Data optimization
-

1.18 MICROSOFT 365 COPILOT

What is Copilot?

Microsoft 365 Copilot combines:

- Large Language Models (LLMs)
- Microsoft Graph
- Microsoft Applications

Applications:

- Excel
- Word
- PowerPoint
- Outlook
- Teams

Copilot in Excel

Capabilities:

1. Natural Language Queries

Examples:

- "Give sales breakdown."

- "Analyze trends."
- "Project impact scenarios."

2. Model Generation

Can:

- Create charts
- Build forecasts
- Identify trends

3. Effortless Analysis

Users no longer need complex formulas.

Copilot in Other Applications

1. Word

- Draft documents
- Rewrite content
- Summarize text

2. PowerPoint

- Create presentations
- Reformat slides
- Generate content

3. Outlook

- Summarize emails
- Draft responses
- Schedule events

4. Teams

- Summarize meetings
- Identify action items
- Create reports

MODULE 1 KEY EXAM POINTS

Definitions

- Generative AI → Creates new synthetic content/data.
- Data Augmentation → Expands datasets artificially.
- Data Preparation → Cleaning and transforming data.
- Database Querying → Retrieving/manipulating data using SQL.
- Data Q&A → Asking questions about data in natural language.

Important Tools

Task	Tools
Data Augmentation	CTGAN, SDV, GauGAN, Imagen, StyleGAN2
Data Preparation	ChatCSV, Tomat.AI
Querying Databases	SQLthroughAI, dbsensei
Data Q&A	Akkio, Powerdrill
BI & Analytics	Cognos, Power BI, Tableau Pulse, ThoughtSpot

Most Important Takeaway

Generative AI is transforming data analytics by enabling users to:

- Generate and augment datasets
 - Prepare and clean data
 - Query databases using natural language
 - Gain insights through conversational Q&A
 - Automate business intelligence and decision-making workflows
-

MODULE 2 – USE OF GENERATIVE AI FOR DATA ANALYTICS

Generative AI for Data Visualization and Storytelling

2.1 GENERATIVE AI FOR DATA INSIGHTS

Overview

Generative AI helps data analysts derive insights from datasets without writing extensive code manually. It can:

- Generate Python scripts automatically.
- Perform statistical analysis.
- Detect missing values.
- Conduct univariate, bivariate, and multivariate analysis.
- Perform feature selection.
- Engineer new features.
- Create visualizations and summaries.

Why Data Insights Matter

Before building predictive models, analysts need to:

- Understand the data.
- Identify relationships among variables.
- Detect missing values and outliers.
- Find influential variables.
- Discover patterns and trends.

Generative AI accelerates these tasks.

Generating Statistical Analysis with AI

Suppose cleaned data is stored in a CSV file.

Example Prompt

Create a Python code to generate the statistical description of cleaned data available in a CSV file.

Generated Python Operations

Using Pandas:

```
import pandas as pd

df = pd.read_csv("data.csv")

description = df.describe(include='all')

print(description)
```

Output Includes

For numerical columns:

- Count
- Mean
- Standard deviation
- Minimum value
- Quartiles (25%, 50%, 75%)
- Maximum value

For categorical columns:

- Number of unique values
 - Most frequent category
 - Frequency of the most common category
-

2.2 ADVANCED DATA ANALYSIS

Generative AI can generate code for:

1. Univariate Analysis

Studies one variable at a time.

Examples:

- Mean
- Median
- Distribution
- Histograms

Purpose:

Understand the characteristics of individual variables.

2. Bivariate Analysis

Examines relationships between two variables.

Examples:

- Correlation
- Regression plots
- Scatter plots

Purpose:

Determine how one variable influence another.

3. Multivariate Analysis

Analyzes multiple variables simultaneously.

Example:

Pairplot

A pairplot contains:

- Scatter plots between every pair of variables.
- Distribution plots along the diagonal.

Purpose:

Understand complex relationships among variables.

Feature Selection

Generative AI can identify the most important predictors.

Example Prompt

Add the aspects of selecting the five best features that fit the target attribute.

Technique Used

Scikit-learn:
SelectKBest

Purpose

Find variables having the strongest relationship with the target variable.

Benefits:

- Improves prediction accuracy.
- Reduces dimensionality.
- Removes irrelevant features.

2.3 FEATURE ENGINEERING

Generative AI can also create new variables.

Example

Polynomial features:

PolynomialFeatures

Purpose:

Capture nonlinear relationships.

Benefits:

- Better model performance.
- More informative features.

Using HAL9 for Data Insights

HAL9 provides AI-assisted analysis.

Dataset Used

Student Performance Dataset

File:

student-mat.csv

Contains:**Academic Features**

- Grades

Demographic Features

- Age
- Gender

Social Features

- Family information
- Alcohol consumption

School Features

- School type

Example 1: Distribution of Student Ages Across Schools**Prompt:**

Find the distribution of student ages across schools.

Result

Graph showing:

School GP:

- 57 students aged 18

School MS:

- 25 students aged 18

Example 2: Missing Value Detection**Prompt:**

Find missing values.

Output:

- Column names.
- Count of missing values.

Result:

No missing values were found.

Example 3: Statistical Summary

Output for numerical variables:

- Count
- Mean
- Standard deviation
- Minimum
- Quartiles
- Maximum

Output for categorical variables:

- Unique values
- Frequency

Example:

School column:

Unique values:

- GP
- MS

Hands-on Lab: Generative AI for Data Insights**Dataset**

Laptop Price Prediction Dataset

File:

URL = https://cf-courses-data.s3.us.cloud-object-storage.appdomain.cloud/IBMDeveloperSkillsNetwork-DA0101EN-Coursera/laptop_pricing_dataset_mod2.csv

Target Variable:

Price

Objective 1: Statistical Description**Prompt**

Write a python code to perform the following actions.

1. Import a dataset from a CSV file.
2. Generate statistical description of all features including object data types.

Python Code

```
import pandas as pd
df = pd.read_csv("data.csv")
description = df.describe(include='all')
print(description)
```

Objective 2: Parameter Visualizations

Two types of variables:

Continuous Variables

- CPU_frequency
- Screen_Size_inch
- Weight_pounds

Visualization

Regression plots

Purpose:

Show relationship with Price.

Categorical Variables

- Category
- GPU
- OS
- CPU_core
- RAM_GB
- Storage_GB_SSD

Visualization

Box plots

Purpose:

Compare Price distributions among categories.

Libraries Used

```
import seaborn as sns  
import matplotlib.pyplot as plt
```

Regression Plot

Shows:

- Trend line
- Strength of relationship
- Positive or negative association

Example:

```
sns.regplot(x='CPU_frequency', y='Price')
```

Box Plot

Displays:

- Median
- Quartiles
- Outliers
- Spread of data

Example:

```
sns.boxplot(x='GPU', y='Price')
```

Objective 3: Evaluating Dependence

Not every variable strongly influences the target.

Important measures:

Correlation Value

Measures strength of relationship.

Range: -1 to +1

Positive Correlation

As one variable increases, the other increases.

Negative Correlation

As one variable increases, the other decreases.

Zero Correlation

No relationship.

Pearson Correlation Coefficient

Calculated using:

```
pearsonr()
```

Example:

```
from scipy.stats import pearsonr
```

p-value

Indicates statistical significance.

Small p-value (<0.05)

Relationship is significant.

Large p-value (>0.05)

Relationship may be due to chance.

Prompt

Evaluate correlation value, Pearson coefficient and p-values for all numerical attributes against Price.

Exclude Price against itself.

Print results in a dataframe.

Output Table

Attribute	Pearson_r	Correlation	p-value
CPU_frequency	0.68	0.68	0.001
RAM_GB	0.74	0.74	0.000
Weight_pounds	-0.42	-0.42	0.02

Objective 4: Grouping and Pivot Tables

Grouping combines multiple variables.

Example variables:

- GPU
- CPU_core
- Price

Pivot Table

Rows:

GPU

Columns:

CPU_core

Values:

Mean Price

Aggregation:

mean()

Example

GPU	2 cores	4 cores	8 cores
Intel	500	800	1200
Nvidia	700	1100	1800

Pcolor Plot

Visual representation of pivot tables.

Color intensity represents:

- Higher values
- Lower values

Implemented using:

<code>plt.pcolor()</code>

Purpose:

Identify patterns visually.

2.4 GENERATIVE AI FOR DATA VISUALIZATION

Generative AI tools can:

- Create charts automatically.
- Generate insights.
- Explain relationships among variables.
- Produce statistical summaries.
- Detect outliers.

Tool 1: Columns AI

AI-powered storytelling platform.

Features

1. Automatic Chart Generation

Produces:

- Histograms
- Bar charts
- Pie charts
- Scatter plots

2. Smart Titles

Automatically generates chart titles.

3. AI Insights

Explains the meaning behind charts.

4. Customization

Allows users to:

- Change colors.
- Modify chart appearance.
- Add annotations.

5. Export Options

Can export:

- PNG
- SVG
- CSV
- JSON

Demonstration with Student Dataset

Gender Distribution

Prompt:

Find total male and female students.

Result:

Bar chart showing:

- Female students = 208
- Male students = 187

Alcohol Consumption by Gender

Visualization:

Pie chart

Variable:

Walc

(Weekly alcohol consumption)

AI generates:

- Chart title.
- Insights.
- Annotations.

Tool 2: Akkio

Uses GPT-4 and Chat Explorer.

Example Dataset

Retail Sales Dataset

Scatter Plot**Prompt:**

Relationship between marketing spend and sales.

Result:

Scatter plot showing dependency.

Bar Chart**Prompt:**

Average sales by area.

Output:

Bar chart.

Correlation Matrix Heatmap**Purpose:**

Understand relationships among variables.

Interpretation

Light colors: Strong correlation.

Dark colors: Weak correlation.

Different colors indicate:

- Positive correlations.
- Negative correlations.

Box Plots

Used for:

Detecting outliers.

Generated for all variables automatically.

Histograms

Show:

- Frequency distribution.
- Spread of values.
- Shape of data.

2.5 GENERATIVE AI FOR CREATING DASHBOARDS

Purpose of Dashboards

Dashboards present information visually and help in:

- Better decision-making.
- Monitoring business performance.
- Identifying trends.
- Communicating insights effectively.

Benefits of Generative AI in Dashboard Creation

Generative AI:

- Automates design choices.
- Reduces coding effort.
- Improves user experience.
- Makes dashboard creation easier.

AI-Based Dashboard Frameworks

1. Dash

Python framework for analytical web applications.

Features:

- Interactive dashboards.
- Integration with LLMs like ChatGPT.
- Highly customizable.

2. ChatGPT

Can generate dashboard code.

Supports:

- Natural language interaction.
- Querying datasets.
- Dashboard development assistance.

3. Tableau AI

Uses:

Einstein Trust Layer

Provides:

- Ethical AI.
- Personalized insights.
- Secure data handling.

Einstein Copilot

Helps with:

- Better visualization recommendations.
- Dashboard automation.
- Productivity enhancement.

4. Akkio

Uses GPT-4.

Generates:

- Scatter plots
- Line charts
- Bar charts

Charts can later be combined into dashboards.

5. ThoughtSpot

Features:

- Natural language search.
- Automated analytics.
- Anomaly detection.
- Intelligent dashboard recommendations.

6. DataSquirrel

Capabilities:

- Auto-cleaning data.
- Auto-analysis.
- Dashboard generation.

Applications:

- HR
- Finance
- Survey analysis
- B2B e-commerce

7. Sisense

AI-powered BI platform.

Provides:

- Automated data preparation.
 - NLP capabilities.
 - Machine learning support.
-

GENERATIVE AI CONSIDERATIONS IN DATA ANALYTICS

2.6 GENERATIVE AI CONSIDERATIONS IN DATA ANALYTICS

Generative AI provides powerful capabilities, but its use requires careful planning and responsible deployment. Data professionals must consider:

1. Data Considerations

2. Model Considerations

3. Ethical Considerations

These considerations ensure:

- Accuracy
- Fairness
- Transparency
- Security
- Responsible AI usage

1. Data Considerations

Generative AI models learn from large datasets.

Importance of Training Data

Model performance depends heavily on:

- Data quality
- Data accuracy

- Representativeness
- Diversity
- Absence of bias

Poor-quality data produces:

- Inaccurate outputs
- Amplified biases
- Unfair decisions

Responsibilities of Data Scientists

They should:

- ✓ Evaluate data quality.
- ✓ Ensure data represents all groups fairly.
- ✓ Remove errors and inconsistencies.
- ✓ Detect and reduce bias.
- ✓ Use diverse datasets.

2. Model Considerations

The selected model determines:

- Explainability
- Interpretability
- Reliability

a). Explainability

Explainability means:

The model can provide insights into how it arrived at a decision.

Example:

A loan approval model should explain why a loan was rejected.

b). Interpretability

Interpretability means:

Humans can understand model outputs easily.

Techniques for Better Interpretability

a). Feature Attribution

Determines which features influence predictions most.

Example:

For loan approval:

- Income = 45%
- Credit score = 35%
- Debt ratio = 20%

b). Partial Dependence Plots (PDP)

Show how a variable affects predictions.

Example:

Relationship between:

- Income and Loan Approval
- Age and Insurance Premium

3. Ethical Considerations

Generative AI can be misused.

Possible misuse:

- **Deepfakes:** Fake videos or images.
- **Manipulating Public Opinion:** False information campaigns.
- **Misinformation:** Generation of fake news.
- **Malicious Content:** Creation of harmful or deceptive information.

Responsibilities of Data Professionals

They should:

- Establish ethical guidelines.

- Monitor misuse.
 - Promote responsible AI practices.
 - Prevent harmful applications.
-

2.7 INDUSTRY-SPECIFIC CONSIDERATIONS

1. Generative AI in Finance

Financial systems handle highly sensitive data.

Examples:

- Banking records
- Credit history
- Transactions
- Investments

A. Data Considerations

Challenges:

- Data privacy
- Regulatory compliance
- Market impact

Solutions

- **Encryption:** Protects sensitive information.
- **Access Control:** Restricts data access.
- **Compliance:** Follow: GDPR and Financial regulations

B. Model Considerations

Models must be:

- Robust
- Explainable
- Resistant to attacks

Adversarial Attacks

Attackers intentionally manipulate inputs to fool models.

Example:

Manipulating transaction data to avoid fraud detection.

Solutions

- **Feature Attribution:** Explains decisions.
- **Partial Dependence Plots:** Improve understanding.
- **Robust Optimization:** Strengthens model resistance.

C. Ethical Considerations

Biased training data may lead to:

- Discriminatory lending
- Unfair loan approvals

Bias Mitigation Techniques

- **Fairness Metrics:** Measure equity among groups.
- **Adversarial Training:** Reduces discrimination.

2. Generative AI in Healthcare

Healthcare data includes:

- Medical records
- Imaging data
- Genetic information
- Clinical trial data

A. Data Considerations

Need:

- High quality
- Representative samples
- Error-free datasets

Compliance Requirements

Must follow:

HIPAA

Protects patient information.

B. Model Considerations

Healthcare AI must be:

- Highly accurate
- Interpretable
- Resistant to attacks

Because wrong predictions may cause:

- Misdiagnosis
- Incorrect treatment
- Patient harm

Privacy Techniques

- **Data Anonymization:** Removes patient identity.
- **Access Control:** Restricts sensitive information.

C. Ethical Considerations

Transparency

Patients should know:

- AI is involved.
- Limitations of AI.
- Potential risks.

Informed Consent

Patients should:

- Approve AI use.
- Access their AI-generated information.

Bias Mitigation

Prevent discrimination across:

- Gender
- Age
- Ethnicity

3. Generative AI in Retail

Retail data includes:

- Purchase history
- Product specifications
- Market trends
- Social media interactions

A. Data Considerations

Need:

- Accurate data
- Effective marketing insights
- Quality product information

Data Augmentation

Generates additional training samples while preserving:

- Patterns
- Distributions

B. Model Considerations

Different tasks require different architectures.

GANs (Generative Adversarial Networks)

Useful for:

- Product image generation

RNNs (Recurrent Neural Networks)

Useful for:

- Purchase prediction
- Sequential customer behavior

Interpretability Techniques

Help explain recommendations.

C. Ethical Considerations

Customer data includes:

- Personal information
- Purchase history

Potential problems:

- Unfair product recommendations
- Privacy violations

Solutions

- Bias mitigation
- Strong data security
- Compliance with privacy regulations
- Customer consent

2.8 CHALLENGES WHILE USING GENERATIVE AI

Challenges are categorized into:

1. Technical Challenges
2. Organizational Challenges
3. Cultural Challenges

1. Technical Challenges

A. Data Quality and Availability

Generative AI needs:

- Large datasets
- High-quality data
- Proper labeling

Problems:

- Limited data
- Sensitive information
- Domain-specific scarcity

B. Model Interpretability

Large models are difficult to understand.

Consequences:

- Hidden biases
- Reduced trust
- Poor explainability

C. AI Hallucinations

Definition

Generation of incorrect or illogical information.

Causes

- Poor training data
- Weak architecture
- Insufficient evaluation

Example

A chatbot inventing non-existent facts.

D. Computational Resources

Training large models requires:

- GPUs
- TPUs
- Cloud infrastructure

Problems:

- High cost
- Resource limitations

E. Lack of Standardization

Generative AI is still evolving.

There are no universal standards for:

- Architectures
- Evaluation metrics
- Training procedures

Making comparison difficult.

2. Organizational Challenges

A. Copyright and Intellectual Property

Concerns:

- Ownership of generated content
- Legal disputes

Solution

Organizations may build:

- Custom AI models

B. Skill Gap

High demand exists for:

- Data Scientists

- Machine Learning Engineers
- AI Specialists

But supply is limited.

C. Integration Challenges

AI integration requires changes to:

- Data pipelines
- Existing systems
- Risk management

D. Change Management

Employees may fear:

- Job displacement
- Privacy risks
- Loss of control

Organizations need:

- Communication
- Training
- Adoption strategies

E. Measuring ROI

Generative AI benefits are often:

- Long-term
- Difficult to quantify

Some benefits:

- Better decision making
- Customer satisfaction
- Innovation

3. Cultural Challenges

A. Risk Aversion

Organizations hesitate to adopt new technology.

B. Data Sharing Problems

Concerns about:

- Proprietary data
- Confidentiality

can reduce collaboration.

C. Trust and Transparency

Generative AI models are often opaque.

Users may question:

- Reliability
- Fairness

D. Continuous Learning

Models require constant:

- Updates
- Monitoring
- Retraining

Organizations must develop a culture of:

- Adaptability
- Data-driven decision making

2.9 RESPONSIBLE AI FOR DATA PROFESSIONALS

Responsible AI is also called:

- Ethical AI
- Trustworthy AI

A. IBM's Five Pillars of Responsible AI

1. Explainability

Users should understand:

- Why decisions are made.
- Which factors influence predictions.

2. Fairness

AI should:

- Avoid discrimination.
- Treat everyone equitably.

3. Robustness

Models should resist:

- Errors
- Attacks
- Unexpected situations

4. Transparency

Organizations should disclose:

- Capabilities
- Limitations
- Decision-making processes

5. Privacy

Protect sensitive information through:

- Differential Privacy
- Federated Learning
- Encryption

B. Benefits of Responsible AI

1. Builds Trust

Users understand AI decisions.

2. Accountability

Organizations remain responsible.

3. Legal Compliance

Supports:

- Existing laws
- Future regulations

4. Social Impact

Promotes:

- Human rights
 - Inclusion
 - Diversity
-

C. Environmental Sustainability

Encourages energy-efficient AI.

1. Long-Term Viability

Avoids:

- Public backlash
- Legal restrictions

2. Responsible AI Practices

- **Transparency:** Explain model capabilities and limitations.
- **Inclusiveness:** Use diverse datasets.

3. Continuous Monitoring

Regularly check:

- Bias

- Drift
- Performance

4. Collaboration

Involve:

- Researchers
- Policymakers
- Ethicists
- Public stakeholders

5. User Empowerment

Allow users to control AI systems.

6. Education and Awareness

Promote understanding of AI ethics.

7. Legal Compliance

Adhere to:

- Existing regulations
- Emerging AI laws

2.10 CASE STUDY: GENERATIVE AI IN HEALTHCARE

1. Data Augmentation

Challenges

- Data quality
- Diversity
- Privacy

Solutions

- **Rigorous Quality Control:** Ensures integrity.
- **Adversarial Training:** Improves diversity.
- **Federated Learning:** Protects privacy.
- **Differential Privacy:** Prevents leakage of sensitive data.

Data Professionals' Approach

- Preserve statistical properties.
- Avoid artificial bias.
- Collaborate with clinicians.

2. Diagnostics

Challenges

Combining:

- Images
- Lab reports
- Patient histories

Solutions

- **Data Harmonization:** Integrates multiple sources.
- **Explainable AI**

Using:

- **Saliency Maps:** Highlight important image regions.
- **LIME:** (Local Interpretable Model-Agnostic Explanations) Explains predictions.
- **Clinical Validation:** Compare AI results with traditional diagnosis.
- **Continuous Monitoring:** Update models as practices evolve.

3. Drug Design

Challenges

- Molecular complexity
- Target specificity
- Side effects

Solutions

Generative AI supports:

- Molecular simulation
- Virtual screening
- Property prediction

Data Professionals' Approach

- Collaborate with pharmaceutical institutions.
- Optimize computational efficiency.

4. Personalized Healthcare

Challenges

- Privacy
- Security
- Fairness

Solutions

- **Patient-Centered Design:** Involve patients during development.
 - **Explainable AI:** Assist doctors rather than replace them.
 - **Human-AI Collaboration:** Maintain clinician oversight.
-

IMPORTANT TERMS

Term	Meaning
Hallucination	Generation of false or illogical information
AI Bias	Systematic unfairness in AI outputs
Fairness	Equal treatment without discrimination
Explainability	Ability to understand AI decisions
Interpretability	Ease of understanding model outputs
Differential Privacy	Protects individual information
Federated Learning	Collaborative learning without sharing raw data
Feature Attribution	Identifies important features
Partial Dependence Plot	Shows variable influence on predictions
Robustness	Resistance to attacks and errors
Transparency	Clear communication about AI decisions

KEY TAKEAWAYS

- ✓ Generative AI requires **data, model, and ethical considerations**.
 - ✓ Industry-specific requirements differ across **finance, healthcare, and retail**.
 - ✓ Major challenges are **technical, organizational, and cultural**.
 - ✓ AI hallucinations, interpretability, and computational cost remain important concerns.
 - ✓ Responsible AI is built on IBM's **five pillars: Explainability, Fairness, Robustness, Transparency, and Privacy**.
 - ✓ Continuous monitoring, collaboration, and legal compliance are essential.
 - ✓ Healthcare applications require strong emphasis on **privacy, explainability, clinical validation, and human-AI collaboration**.
-

MODULE 3: FINAL PROJECT AND EXAM

3.1 GUIDED PRACTICE PROJECT OVERVIEW

Scenario

You are working as a **Data Analyst** for a healthcare consultancy firm.

The organization studies the **World Happiness Report** and wants to determine:

- Whether demographic characteristics affect happiness.
- Whether regional differences influence quality of life.
- Whether economic factors lead to better living standards.

Generative AI is used to automatically generate Python code for:

1. Data preparation
 2. Data analysis
 3. Data visualization
 4. Dashboard creation
 5. Storytelling
-

3.2 DATASET USED

World Happiness Report (2016)

Source:

- Kaggle (Public Domain)

Dataset URL:

URL = "https://cf-courses-data.s3.us.cloud-object-storage.appdomain.cloud/IBMSkillsNetwork-AI0272EN-SkillsNetwork/labs/dataset/2016.csv"

Dataset Attributes

Variable	Description
Country	Country name
Region	Region to which the country belongs
Happiness Rank	Rank based on happiness score
Happiness Score	Overall happiness measure
Lower Confidence Interval	Lower confidence limit
Upper Confidence Interval	Upper confidence limit
Economy (GDP per Capita)	Economic contribution to happiness
Family	Family support contribution
Health (Life Expectancy)	Health contribution
Freedom	Freedom contribution
Trust (Government Corruption)	Trust contribution
Generosity	Generosity contribution
Dystopia Residual	Unexplained happiness component

3.3 PROJECT WORKFLOW

The project consists of:



↓
Dashboarding
↓
Storytelling

1. Data Preparation

Step 1: Load Dataset

Task: Read CSV file into a Pandas DataFrame and display first 5 rows.

Prompt: Write a Python code that reads the CSV file into a pandas dataframe and prints the first five rows.

Code

```
import pandas as pd
df = pd.read_csv("dataset.csv")
print(df.head())
```

Step 2: Check Data Types

Incorrect datatypes can produce wrong analysis.

Prompt: Write a python code to check the data types of all columns.

Code

```
print(df.dtypes)
```

Step 3: Change Data Types

Tasks:

Remove Whitespaces

```
df["Country"] = df["Country"].str.strip()
```

Replace Empty Strings with NaN

```
import numpy as np
df.replace({"": np.nan}, inplace=True)
```

Convert Columns to Numeric

```
numeric_cols = [  
'Happiness Rank',  
'Happiness Score',  
'Lower Confidence Interval',  
'Upper Confidence Interval',  
'Economy (GDP per Capita)',  
'Family',  
'Health (Life Expectancy)',  
'Freedom',  
'Trust (Government Corruption)',  
'Generosity',  
'Dystopia Residual'  
)  
  
for col in numeric_cols:  
    df[col] = pd.to_numeric(df[col], errors='coerce')
```

Step 4: Handle Missing Values

Identify Missing Values

```
print(df.isnull().sum())
```

Fill Missing Values with Mean

```
for col in numeric_cols:  
    df[col].fillna(df[col].mean(), inplace=True)
```

2. Database Querying (Optional)

Count Rows for Each Country

```
SELECT Country,  
COUNT(*) AS TotalRows  
FROM happiness  
GROUP BY Country;
```

Region-wise Average Happiness Score

```
SELECT Region,  
AVG([Happiness Score]) AS Avg_Happiness  
FROM happiness  
GROUP BY Region;
```

3. Exploratory Data Analysis (EDA)

Task 1: Top 10 Countries

Find:

- GDP per Capita
- Healthy Life Expectancy

Select Top 10 Countries

```
top10 = df.nlargest(10, 'Happiness Score')
```

Columns Required

```
top10[['Country',  
      'Economy (GDP per Capita)',  
      'Health (Life Expectancy)']]
```

Visualization 1: Bar Chart (fig1)

Using Plotly

```
import plotly.express as px  
fig1 = px.bar(  
    top10,  
    x='Country',  
    y=['Economy (GDP per Capita)',  
      'Health (Life Expectancy)'],  
    barmode='group',  
    title='Top 10 Countries'  
)  
fig1.show()
```

Task 2: Correlation Analysis

Columns:

- Economy (GDP per Capita)
- Family
- Health (Life Expectancy)
- Freedom
- Trust (Government Corruption)
- Generosity
- Happiness Score

Create Subset

```
subset = df[[  
'Economy (GDP per Capita)',  
'Family',  
'Health (Life Expectancy)',  
'Freedom',  
'Trust (Government Corruption)',  
'Generosity',  
'Happiness Score'  
]]
```

Correlation Matrix

```
corr = subset.corr()
```

Visualization 2: Heatmap (fig2)

```
fig2 = px.imshow(  
    corr,  
    width=800,  
    height=600,  
    text_auto=True,  
    title="Correlation Heatmap"  
)  
fig2.show()
```

Task 3: GDP vs Happiness Score

Study relationship between:

- GDP per Capita
- Happiness Score

Grouped by:

- Region

Visualization 3: Scatter Plot (fig3)

```
fig3 = px.scatter(  
    df,  
    x='Economy (GDP per Capita)',  
    y='Happiness Score',  
    color='Region',  
    title='GDP vs Happiness Score'  
)  
fig3.show()
```

Interpretation

Higher GDP generally leads to higher happiness.

Regions show distinct clusters.

Task 4: Happiness Score by Region

Visualization 4: Pie Chart (fig4)

```
region_data = df.groupby('Region')['Happiness Score'].mean().reset_index()  
fig4 = px.pie(  
    region_data,  
    values='Happiness Score',  
    names='Region',  
    title='Happiness Score by Region'  
)  
fig4.show()
```

Task 5: World Map

Display:

- GDP per Capita

Tooltip:

- Healthy Life Expectancy

Visualization 5: Choropleth Map (fig5)

```
fig5 = px.choropleth(  
    df,  
    locations='Country',  
    locationmode='country names',  
    color='Economy (GDP per Capita)',  
    hover_data=['Health (Life Expectancy)'],  
    title='GDP per Capita Across Countries'  
)  
fig5.show()
```

4. Dashboarding

At least four visualizations are required.

Possible figures:

```
fig1 → Bar Chart  
fig2 → Heatmap  
fig3 → Scatter Plot  
fig4 → Pie Chart  
fig5 → World Map
```

Writing Dashboard to HTML

```
from plotly.subplots import make_subplots
```

```
import plotly.graph_objects as go
```

Alternative:

```
with open("dashboard.html", "w") as f:  
    f.write(fig1.to_html(full_html=False,  
                        include_plotlyjs='cdn'))  
  
    f.write(fig2.to_html(full_html=False,  
                        include_plotlyjs=False))  
  
    f.write(fig3.to_html(full_html=False,  
                        include_plotlyjs=False))  
  
    f.write(fig4.to_html(full_html=False,  
                        include_plotlyjs=False))
```

Output

Creates:

dashboard.html

containing multiple interactive visualizations.

5. Storytelling and Narrative

A dashboard alone is not enough.

Data analysts should explain the insights.

Heatmap Insights

- GDP has strong positive correlation with happiness.
- Health and Family support are also major contributors.
- Generosity and trust have weaker correlations.

Scatter Plot Insights

- Countries with higher GDP generally show higher happiness scores.
- Regional differences exist.
- Wealth is a major predictor of happiness.

Pie Chart Insights

Different regions contribute differently to global happiness.

Some regions consistently report higher happiness levels.

Map Insights

Developed countries tend to have:

- Higher GDP
- Better life expectancy
- Greater happiness

6. Business Narrative

The World Happiness Report dashboard reveals that economic prosperity, health, and social support are strong drivers of happiness. Correlation analysis indicates that GDP per capita and life expectancy are highly associated with happiness scores. Scatter plot analysis further confirms that countries with stronger economies generally experience greater happiness levels, though regional differences are evident. The geographic map demonstrates that nations with higher GDP also tend to have better life expectancy. These insights suggest that improving economic conditions and healthcare systems can significantly enhance quality of life across countries.

7. Test Environment Setup

Install Libraries

```
%pip install seaborn
```

Download Dataset

```
import requests
response = requests.get(URL)
if response.status_code == 200:
```

```
with open("dataset.csv", 'wb') as f:  
    f.write(response.content)
```

3.4 OVERALL PROJECT FLOW

```
graph TD; A[Load Dataset] --> B[Check Datatypes]; B --> C[Clean Data]; C --> D[Handle Missing Values]; D --> E[EDA]; E --> F[Bar Chart]; F --> G[Heatmap]; G --> H[Scatter Plot]; H --> I[Pie Chart]; I --> J[Map]; J --> K[Dashboard]; K --> L[Narrative/Storytelling];
```

Load Dataset
↓
Check Datatypes
↓
Clean Data
↓
Handle Missing Values
↓
EDA
↓
Bar Chart
↓
Heatmap
↓
Scatter Plot
↓
Pie Chart
↓
Map
↓
Dashboard
↓
Narrative/Storytelling

3.5 COURSE WRAP-UP: ENHANCING DATA ANALYTICS WITH GENERATIVE AI

According to Anaconda surveys:

Most Time-Consuming Tasks

Task	Percentage
Data preparation and cleaning	37.75%
Visualization	12.99%
Reports and presentations	6.2%

Generative AI Improves

Data Preparation

- Missing value detection
- Standardization
- Normalization
- Feature selection

Data Analysis

- Statistical summaries
- Correlation analysis
- Synthetic data generation

Visualization

- Charts
- Heatmaps
- Dashboards

Storytelling

- Automatic report generation
- Presentation narratives
- Executive summaries

Ethical Concerns

Organizations should ensure:

- **Fairness:** Avoid bias.
- **Transparency:** Explain decisions clearly.
- **Privacy:** Protect sensitive data.
- **Explainability:** Models should be understandable.
- **Sustainability:** Use AI responsibly.

KEY TAKEAWAY

Generative AI acts as a productivity accelerator for data analysts by automating data preparation, analysis, visualization, dashboarding, and storytelling, allowing analysts to focus more on deriving insights and making strategic decisions.
